



TANIMA HOSSAIN

Software Engineer | AI Researcher

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🎓 EDUCATION

MSc in Computer Science & Engineering,
Daffodil International University [🔗](#)
Major: Data Science

01/2025 - Present | Dhaka, Bangladesh

BSc in Computer Science & Engineering,
Daffodil International University (DIU) [🔗](#)
CGPA: 3.83 out of 4.00

01/2018 - 01/2022 | Dhaka, Bangladesh

💼 WORK EXPERIENCE

AI Research Engineer, InfinitiBit Ltd [🔗](#)

03/2025 - Present | Dhaka, Bangladesh

- Engineered the first Rust-based open source agentic AI framework, **GraphBit** [🔗](#), with a Python wrapper for speed, usability, and efficiency.
- Implemented multiple LLM-providers support for Graphbit, such as AzureOpenAI, MistralAI, Anthropic, ReplicateAI.
- Integrated Multiple VectorDB such as ChromaDB, FAISS, Weaviate, ElasticSearch, Pinecone, Milvus, MariaDB, Qdrant with Graphbit.
- Developed multiple AI agents with tool-calling ability using Graphbit, such as chatbot, resume ranker etc.
- Researching Context Engineering to Improve **GraphBit's** context management and output.
- Implementing a cloud platform for **GraphBit**, where people can publish and deploy AI agents as a service.

Software Engineer, Cefalo Bangladesh Ltd. [🔗](#)

02/2022 - 08/2024 | Dhaka, Bangladesh

- Worked with a team to build the best web-based gamified math learning platform, the **House of Math** [🔗](#).
- Led system design and mentored junior developers, ensuring timely, high-quality feature delivery.
- Enhanced user experience by designing a **Graph Algorithm** for personalizing learning paths over 150+ topics.
- Upheld coding standards and mitigated potential issues by conducting over 80 **Code Reviews** of peer work.
- Minimized business impact by resolving over 100 **Production Issues**, ensuring smooth operation.
- Ensured comprehensive feature coverage by writing 30+ **End-to-End** and **Unit** tests using **Playwright** and Jest.

Problem Solving and Teaching Experience

- Competitive Programming Bootcamp for Female Students [🔗](#) [12/2020 - 02/2021] | Organizer: EMK Center [🔗](#)
- Competitive Programming Trainer [02/2022 - 07/2022] | Organizer: Canadian University of Bangladesh [🔗](#)
- Problem-Solving Trainer [05/2019 to 12/2021] | Organizer: ACM of Daffodil International University [🔗](#)
- Teaching Assistant [Spring 2019 - Spring 2021] | Courses: Problem Solving, Data Structures, Algorithms | Institute: Daffodil International University [🔗](#)

🏆 HONOURS AND AWARDS

Champion, DIU Inter Department Math Olympiad 2019 [🔗](#), Onsite Individual

Awarded the champion's trophy by the General Secretary of Bangladesh Math Olympiad Committee for being the top scorer among 300+ participants, organized by Daffodil International University.

Champion, DIU Intra University Girls Programming Contest, Spring 2019 [🔗](#), Onsite Individual

I was awarded the champion's trophy by Mr. Shahriar Manzoor, judge of ACM-ICPC World Finals in the contest organized by Daffodil International University.

Academic Waiver for Brilliant Result

I received a 50% tuition fee waiver scholarship from Daffodil International University in the Fall 2019 and Spring 2021 semesters, for Brilliant Academic Results.

Runner-up, DIU Intra University Programming Contest 2021 [🔗](#), Onsite Individual

Achieved 1st runner-up rank in this contest organized by Daffodil International University.

Runner-up, DIU Take-off Programming Contests Summer 2018 [🔗](#), Onsite Individual

Achieved 1st Runner-up rank in this contest targeted towards freshmen of Daffodil International University.

Runner-up, Inter University Girls' Programming Contest Season 2 [🔗](#), Onsite Team contest

Participated as DIU_Heuristic, achieving Runner-up in the contest organized by North South University.

2nd Runner-up, Ada Lovelace National Girls' Programming Contest 2020 [🔗](#), Onsite Team Contest

Participated as DIU_Heuristic, achieving 2nd Runner-up in this contest organized by University of Asia Pacific.

SEMINARS

Ignite Programming Passion (Keynote Speaker): Was Invited to inspire the female students of Daffodil International University to uplift their competitive programming passion. [Promotion [↗](#), Highlights & photos [↗](#)]

Ready, Set, Code (Keynote Speaker): Invited to introduce competitive programming to the freshmen and sophomores of Daffodil International University. [Promotion [↗](#), Highlights & photos [↗](#)]

Ice-Breaking Session (Guest Speaker): Invited for an interactive session to ignite the programming passion of students of the Canadian University of Bangladesh as their programming trainer. [Promotion [↗](#)]

Competitive Programming and Career Excellence (Host): Led an insightful session offering guidance on preparation and achieving career success through competitive problem-solving. [Promotion [↗](#), Event [↗](#)]

Women in Tech (Participant): Invited to join a select forum of highly accomplished female professionals from diverse fields by Optimizely to share career insights and collaborate on strategies for industry advancement. [Event [↗](#)]

MANAGEMENT AND LEADERSHIP

Managed Competitive Programming Community in DIU 01/2020 - 12/2021

As the top competitive programmer for 2 consecutive years, I have maintained the Ranklist, introduced monthly programmers meet, and arranged several programming classes, contests, and programming camps for DIU ACM [↗](#).

</> SKILLS

Technical Skills: C/C++, Rust, TypeScript, JavaScript, Python, C#, Java, Express.js, Node.js, React.js, Next.js, Android, .Net 6, MySQL, PostgreSQL, Sequelize ORM, Knex.js, HTML, CSS, TailwindCSS, Git, Docker, Postman, Playwright, Sanity.io, Stripe.

Soft Skills: Software Development, Problem solving, Public speaking, Community management, Teaching

Others: Click Up, Jira, Google Workspace, Microsoft Office (Word, Excel, PowerPoint), Documentation

LANGUAGES

English: IELTS academic overall band score: **7** | CEFR: C1 | Listening: 8, Reading: 6, Writing: 6.5, Speaking: 7.5

COMPETITIVE PROGRAMMING AND PROBLEM SOLVING

Problem Solving

Solved over 3500 data structures and algorithmic problems in various **online judges** [↗](#).

Participated in numerous onsite and online programming contests representing my university. Some notables are -

Individual Contest

- **Topper** [↗](#) in DIU Programmers Individual Rank list - 2020 & 2021
- **Candidate Master** at Codeforces [↗](#), Max Rating: 1974 | Best Ranks: Div. 1 [↗](#) : 1028/10630 (Top 10%), Div. 2 [↗](#) : 83/10990 (Top 1%), Div. 3 [↗](#) : 197/4974 (Top 4%).
- **Knight** at LeetCode [↗](#), Max Rating: 1913 (Top 5%) | Best Rank: biweekly 93 [↗](#) : 257/17080 (Top 2%).
- **4 star** at Codechef [↗](#), Max Rating: 1925 | Best Rank: Div. 2 [↗](#) : 44/5465 (Top 1%).
- **S ranked** at Toph [↗](#), Max Rating: 1879 | Best rank: 16th December [↗](#) : 9/321 (Top 3%).
- Google CodeJam 2020 | Round 1B, Rank: 1392 | Round 2, Rank: 3291

Team Contest

- **Champion** among all-girls team, Team: DIU_Heuristic, ICPC Dhaka Regional 2018 [↗](#)
- **Champion** among all-girls team, Team: DIU_Heuristic, BUP IUPC 2019 [↗](#)
- **5th**, Team: DIU_Heuristic, Ada Lovelace National Girls' Programming Contest 2021 [↗](#)
- **5th**, Team: DIU_Heuristic, National Girls' Programming Contest 2021 [↗](#)
- **12th**, Team: DIU_Heuristic, National Girls' Programming Contest 2019 [↗](#)
- **13th**, Team: DIU_Heuristic, DIU NGPC 2018 [↗](#)
- Top 40%, Team: DIU_TrOjAnZ, ICPC Dhaka Regional 2020 [↗](#)
- Top 40%, Team: DIU_TrOjAnZ, MIST NCPC 2020 [↗](#)
- Top 40%, Team: DIU_TrOjAnZ, ICPC Dhaka Regional 2019 [↗](#)

PROBLEM SETTER & JUDGE

Authored numerous data structures and algorithms (DSA) problems and acted as a judge in multiple programming contests.

- **Judging Director & Problem Setter**, DIU Take-off Programming Contest, Spring 2020 [↗](#)
- **Assistant Judging Director & Problem Setter**, STEM Fiesta Girls Programming Contest [↗](#)
- **Problem Setter & Judge**, Code for Athoy - Contest for Saving a Life [↗](#)
- **Problem Setter & Judge**, Around BD Programming Contest 2020 [↗](#)
- **Special Judge**, DIU Take-off Programming Contest, Spring 2022 [↗](#)
- **Problem Setter & Judge**, DIU Take-off Programming Contest from Spring 2019 to Fall 2019